

Sana Nazir

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PhD researcher developing high-order, hardware-agnostic, performance-portable CFD solvers in C++/Kokkos.

EDUCATION

PURDUE UNIVERSITY

PHD AERONAUTICS &
ASTRONAUTICS
Dec 2027 | USA

MSC AERONAUTICS &
ASTRONAUTICS
Dec 2022 | USA

MSC COMPUTER SCIENCE
May 2018 | USA

BSC COMPUTER SCIENCE
Lahore University of
Management Sciences |
PAKISTAN

COURSEWORK

CONCENTRATIONS

Aerodynamics

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Propulsion

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Compilers

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Networks & Operating Systems

ACADEMIC PROJECTS

2025 | Temperature-Dependent
Modeling of N₂ Plasma Emission
Spectra (Python)

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2024 | Identification of Vortex
Stretching Events in Turbulent
Channel Flow (JHTDB)

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2024 | Hypersonic DSMC Study of
Hollow-Cylinder-Flare (SPARTA)

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2022 | Hydrogen Powered
Micro-Gas-Turbine

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2022 | Ultra High Bypass Ratio
Geared Turbofan Engine Analysis

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2018 | MapReduce Word Count
(C++, OpenMP, MPI)

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2017 | GCC Compiler Support for
Analyzing Uninitialized Local/Global
Variables (C)

EXPERIENCE

PURDUE UNIVERSITY | PHD STUDENT | GRADUATE STAFF

- Jan 2023 - present | Teaching Assistant Aeronautics & Astronautics

INTEL | SYSTEMS/ COMPILER ENGINEER

- Aug 2019 - July 2021 | Software Engineer IOTG - Compiler Architecture
- Oct 2018 - Aug 2019 | Software Engineer IAGS - Systems Software
- May 2017 - Aug 2017 | Graduate Software Engineering Intern
- May 2016 - Aug 2016 | Graduate Software Engineering Intern

SELECT PROJECTS

PRESENT - PURDUE | HIGH-ORDER CFD FRAMEWORK IN C++/KOKKOS

Developing a hardware-agnostic high-order CFD framework using explicit and compact finite-difference schemes on structured and curvilinear meshes, targeting performance portability across CPU/GPU architectures with OpenMP, CUDA, and HIP.

- Presented @ HPSF Conference 2026; AIAA SCITECH 2026.
- Accepted @ AIAA SCITECH 2027

PRESENT - PURDUE | AEROSPACE SCIENCES LAB | SB LI AT MACH 2.0

Developing a color schlieren visualization technique for experimental investigation of swept oblique shock-wave/boundary-layer interactions (SB LI) at Mach 2.0.

- Presented @ AIAA SCITECH 2025.

PRESENT - PURDUE | PLASMA & ELECTRIC DISCHARGES COURSEBOOK

Co-authoring a comprehensive graduate-level textbook on Plasmas and Electric Discharges.

INTEL

2019 | PLAIDML EXECUTION PROVIDER FOR ONNX RUNTIME

Created the interface to enable ONNX Runtime with PlaidML as the execution provider, by extending the Tile eDSL to include ONNX operations.

2018 | MOHAWK RIVER AMBIENT COMPUTE PROTOTYPE

Built a Face Recognition Module utilizing a 360 camera device for a dual-OS shared peripheral system.

- Prototype showcased @ COMPUTEX 2019, INTEL DTTC 2019, INTEL SWPC 2019, INTEL Demo Day 2020

2017 | FEDERATED LEARNING DEMONSTRATION

Demonstrated Federated Learning using a two-layer CNN on the MNIST dataset. Setup with Intel NUC devices and multiple connected smartphones. Built using TensorFlow and Python.

- Presented as live demo and poster @ Intel "Showcase Your Amazing" 2017.

2016 | TELECONFERENCING & IROBOT OBSTACLE AVOIDANCE

Developed a 360° teleconferencing application with automatic speaker detection and an iRobot obstacle-avoidance system using Intel RealSense depth cameras and computer vision.

- Prototype showcased at Intel DTTC 2017.
- Received Significant Effort and Achievement Award for measurable work-group impact.